L6.3: Examples of finding bases for the kernel and image of a linear transformation

Transcript Language

English

Hello, and welcome to the Maths 2 component of the online B.Sc. program on data science

and programming. In this video, we are going to continue from the previous video where

we discussed the theory about the relationship between images and kernels for a linear transformation

with the column space and null space of the matrix that we obtain after we fix ordered

basis for the domain and the codomain.

So, we are going to do a bunch of examples, where we implement what we have seen earlier

and we will find basis for the kernel and the image using the fact that we can find

basis for the column space and the null space using row reduction.

Let us just recall first what I just said. So, we have defined in the previous video

what is the kernel of a linear transformation. For a linear transformation f V to W, let

A be the matrix corresponding to f after choosing ordered basis, beta which is v1, v2, vn and

gamma which is w1, w2, wm for V and W, respectively. How do I get this matrix?

Well, you apply f on vi and express that in terms of the w1, w2, wm and then the coefficients

that you get, you put all of them together and you get your m by n matrix. Once we choose

the basis v1, v2, vn we have this isomorphism Rn to V. And what is the isomorphism. If you

have a column vector in Rn, c1, c2, cn then that corresponds to the vector summation cjvj.

You take the corresponding linear combination that is the isomorphism from Rn to V.

And in the previous video, what we have discussed is that, if you consider the null space of

the matrix A, which is a subspace of Rn, then under this above isomorphism, that gives you

an isomorphism with the kernel of f. This was the import of the first part of the previous

video. So, in particular, if we can get a basis of the null space of A, then using this

isomorphism, we can translate it to a basis of the kernel of f. That is one of the things

we will do in this video for concrete examples.

Similarly, we define the image of a linear transformation. And once again, we can choose

ordered basis and then we get the matrix A. And now we have an isomorphism from Rm to

W. This is because of the choice of ordered bases w1 to wm. So, if you have a vector d1,

d2, dm, corresponding to that vector, you have the linear combination summation djwj.

The main point here is that, if you look at the column space of A, which notice is a subspace

of Rm, then under this isomorphism, its image will be exactly the image of f. So, that is what we discussed in the previous video when we talked about the relation between image

and column space.

And in particular, what this will do is, it will tell you that if you can get a basis

of the column space, then by using this isomorphism, you can also get a basis for the image of

f and we will implement this in some examples now.

So, let us do this example. Consider the linear transformation from R4 to R3 defined by T

of x1, x2, x3, x4, which is 2 times x1 plus 4 times x2 plus 6 times x3 plus 8x4, x1 plus

3x2, 5x4 and x1 plus x2 plus 6x3 plus 3x1. So, the first thing we have to do is choose

ordered basis. So, let us choose the ordered basis to be the standard basis for R4 and

R3. So, once we do that, what is a corresponding matrix where it is exactly the coefficients

of, that are appearing this in this linear transformation. The first row is the coefficients

appearing the first equation. So, those are 2, 4, 6, 8 and then a 1, 3, 0, 5 and then

1, 1, 6, 3.

So, let us row reduce. So, if you row reduce, well, what do I want to do? Maybe before I

row reduce, let us ask, we want to get the basis for the image and the kernel of f of

T and to do that, we have to get basis for the null space and the column space. And we

know how to do that that is why row reduction. So, that is why we are row reducing. So, if

you row reduce, well, we have a 2 in 1, 1 place, so we divide by 2. So, we get this

matrix 1, 2, 3, 4; 1, 3, 0, 5; 1, 1, 6, 3.

Sweep out the first column below the one. So, that is R2 minus R1 and R3 minus R1. So,

if we do that, we get the first row is unchanged 1, 2, 3, 4 and in the second row, we will

get 0, 1, minus 3, 1 and in the third row, we will get 0, minus 1, 3 and minus 1. So,

already you can see that the third row is a multiple of the second row. So, it is going

to get knocked out in one more step.

Well, in the one, so we are done with the first column. So, now we look at the 2, 2th

entry. It is conveniently 1. And so we can knock out the entry below that. So, that in

fact knocks out the entire third row by doing R3 plus R2. So, now this is already in pretty

good shape. But let us do one more step and convert it into reduced row echelon form.

So, to do that, we do R1 minus 2R2 and then that gives us 1, 0, 9 and 2 in the first row,

and then 0, 1, minus 3, 1, 0, 0, 0, 0.

So, that is the reduced row echelon form, so 1, 0, 9, 2, 0, 1, minus 3, 1, 0, 0, 0,

0. And from here, we can read off the basis for the null space and the column space. So,

let us look at first what are the non-pivot and the pivot element elements. So, the pivot

elements are in the 1, 1 and 2, 2 place. So, the non-pivot columns which will correspond

to independent variables are in the third and fourth columns. And that tells us that

x3 and x4 are the independent variables, x1 and x2 are dependent variables. And if we put x3 is s and x4 is t, then we can use the first row and the second row to get the corresponding

equations. And from there, we will get that x1 is minus 9s minus 2t and x2 is 3s minus

t.

So, just to be clear, this is in order to get the basis for the null space. So, for

that, we have to look at the homogeneous equation. So, that is what, that is how we are getting

these equations for x1 and x2. And what that means is, if you substitute s is 1 and t 0

or s is 0 and t is 1 this give you two basis vectors and that will contribute the basis

for the null space. So, if we do that, what do we get? So, if s is 1 and t 0, we get minus

9, comma 9, comma 1, comma 0. And if s is 0 and t is 1, we get minus 2 comma, minus

1 comma, 0 comma, 1. So, that is a basis for the null space.

And since we have chosen beta to be the standard ordered basis, the basis for the kernel of

t is exactly the same. So, it is minus 9 comma 3 comma 1 comma 0. How do I get that? That is because we take the linear combination of the corresponding ordered basis. And in

this case, that is minus 9 times e1 plus 3 times e2 plus 1 times e3 plus 0 times e4,

which is the same as the vector you started with and similarly for the second one. So,

we have got a basis for the kernel of t.

To get a basis for the column space, note that the pivot columns are the first and second

columns. So, that means from the original matrix, if we look at the first two columns,

those will be a basis for the column space. So, that is the vectors 2, 1, 1, and 4, 3,

1. So, just to remind you, here is the matrix. So, the first two columns are, in the first

column we had 2, 1, 1, in the second column we had 4, 3, 1. That is a basis for the column

space.

So, that tells us that again, that since we choose the standard basis, the standard ordered

basis, the image of t also has the same basis. So, 2 times e1 plus 1 times e2 plus 1 times e3 which is 2, 1, 1 and 4 times e1 plus 3 times e2 plus 1 times e3 which is 4, 3, 1.

So, I hope this example demonstrated how to implement what we studied in the previous

video.

Let us do one more example, where we will use basis which are not the standard basis

and so we will have to actually compute what the, what those linear combinations. So, let

us look at V is R2 and W is this subspace of R3, x, y, z, such that x plus y plus z

is 0. So, let us look at respective ordered pieces. So, beta is 1, 1 and 1, minus 1 for

R2. So, notice I am not using the standard ordered basis here, but the ordered basis

1, 1, 1, minus 1. So, for gamma, let us choose the ordered basis minus 1, 1, 0 and minus

1, 0, 1.

Let us look at the linear transformation, T of x, y is 0 comma x plus 2y comma minus

x minus 2y. So, of course, we know this is a linear transformation from R2 to R3. How do I know it is a linear transformation to W? Well, if you look at the image, that is

inside W, because the vectors 0 comma x plus 2y comma minus x minus 2y satisfies that the

sum of these three is 0. So, 0 plus x plus 2y plus minus x minus 2y is indeed 0. So,

this is a linear transformation from V to W.

So, now let us write down the corresponding matrix. So, here, we will have to do some

work. So, here, we have to take the first vector in the basis, which is 1 comma 1, and

then apply T to that, if you apply T to that, you get, so T of 1, 1 is a 0 comma 3 comma

minus 3, and then we have to express that in terms of the basis gamma.

So, 0 comma 3 comma minus 3 is 3 times minus 1 comma 1 comma 0 plus minus 3 times minus

1 comma 0 comma 1. This is something you can take. And that gives us our first column,

which is 3, minus 3.

Similarly, for the second basis vector, we have T of 1 comma minus 1, so that will give us 1 minus 2, which is minus 1 and then minus 1 plus 2, which is 1. So, we get 0 comma minus

1 comma 1 and again it is a check that this leads to the column minus 1, 1 in the second

column. So, this is our corresponding matrix. This is something we have done this in the

previous videos. So, I will encourage you to check this.

So, now, we want to get the basis for the column space and the null space of this matrix.

So, to do that, we will put it into reduced row echelon form. This is a particularly easy

matrix. So, you divide the first row by 3. So, you get 1, minus one-third, minus 3, 1

and then sweep out this is actually clearly the second row is a multiple of the first.

So, you will get 1, minus 1 by 3, 0, 0. So, this is in reduced row echelon form already.

So, now, let us check out what are the pivot and non-pivot columns. So, the pivot column

is the first column. The non-pivot column is the second column. So, x1 is the corresponding

variables x1 is the dependent variable, x2 is the independent variable. And for the homogeneous equation, x1 satisfies, so x1 minus one-third x2 is 0, so x1 is equal to one-third times

x2.

So, if you put x2 is t, then x1 is going to be given by t by 3. So, a basis for the null

space is we can obtain that by putting t is 1. So, if you do that, we get the singleton

set, which is a basis namely, one-third comma 1. So, this is a basis for the null space

of A. Now, from here, I want to get the basis for the kernel of t. So, remember, we have

chosen the basis 1 comma 1 and minus 1 comma 1, I just, sorry, 1 comma minus 1, beta is

1 comma 1, 1 comma minus 1. So, now, we have to take that linear combination one-third

times 1 comma 1 plus 1 times 1 comma minus 1. So, if you evaluate that, that gives us

4 by 3 comma minus 2 by 3. So, that is a basis for kernel of t.

Now, of course, you can simplify this basis. So, you can pull out the one-third or you

can pull out the two-third if you want and instead of this basis you can write this as

2 comma minus 1, so that is another basis for kernel of t. So, similarly, let us go

about what happens to the column space. Let us see what happens to the column space of

A. So, that is, so to do that, we have to see what is the pivot column. So, the pivot

column was the first column. So, that means the first column of the original matrix which

was 3 minus 3 is a basis for the column space.

So, now, we have 3 comma minus 3. And so what we should do is take the linear combination

3 times the first basis vector in gamma plus minus 3 times the second basis vector in comma.

So, if you do that, we get 3 times minus 1, 1, 0 plus minus 3 times minus 1 comma 0 comma

1, and that gives us 0 comma 3 comma minus 3. So, this is a basis for the image of the

linear transformation t. So, I hope this example has again shed more light on the calculations

that we, the theory that we saw in the previous video.

So, let us end this video with the rank nullity theorem for linear transformations. We have actually studied this theorem for matrices. So, now, because we have this very nice connection

between the null spaces and the kernel and image and the column space, we can convert

the rank nullity theorem for matrices into a theorem for linear transformations. So,

suppose T V to W is a linear transformation then the rank of T is the dimension of the

image of T. This is the definition. And the nullity of T is the dimension of kernel of

T.

So, based on what we have studied earlier, notice that in order to compute the rank,

we can choose ordered basis for the domain and codomain and using that we can get a matrix

and the rank of T will be exactly the rank of that matrix. And similarly, the nullity

of T will be exactly the nullity of that matrix.

So, if we reinterpret the rank nullity theorem for matrices, we will thus obtain rank of

T plus nullity of T is dimension of V. Why dimension of V, because remember that what we will get is rank of the matrix plus nullity of the matrix is equal to n. What was n? N

was the number of columns in that matrix. But what is that number of columns? That number

of columns is corresponding to the dimension of V. So, hence we get dimension of V here.

So, I guess that ends this video. Thank you.